MIL-P-17303C(NAVY)
14 October 1955
SUPERSEDING
MIL-P-17303B(NAVY)
9 March 1953

MILITARY SPECIFICATION

PACKING MATERIALS,

PLASTIC METALLIC AND PLASTIC NONMETALLIC

All interested Bureaus of the Department of the Navy have concurred in the use of this specification.

1. SCOPE

1.1 Scope. - This specification covers plastic metallic packings for rotary and reciprocating pumps handling steam, water, gas and oil; and plastic nonmetallic packings for high and low pressure valve stem and reciprocating rod saturated and superheated stem service for machinery on Naval vessels.

1.2 Classification. - The packing shall be of the following classes and types, as specified (see 6.1):

Class I - Plastic metallic, lead base:
- Type A - Bulk form, symbol 1431.
- Type B - Coil form, with or without cotton jacket, symbol 1433.
- Type C - Coil form, metal foil jacket, symbol 1434.
- Type D - Coil form, braided metal jacket, symbol 1432.
- Type E - Coil form, with or without cotton jacket, (gasoline service), symbol 1437.

Class II - Plastic nonmetallic:
- ~~Type B - Coil form, cotton jacket, symbol 1106.~~
- Type C - Coil form, asbestos and nickel-copper wire jacket, symbol 1108.
- Type D - Coil form, with or without cotton jacket, (gasoline service), symbol 1109.
- Type E - Coil form, asbestos and nickel-chromium-iron alloy wire jacket, symbol 1111.

Class III - Plastic metallic, copper base:
- Type B - Coil form, cotton jacket, symbol 1439.

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2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-591 - Boxes, Fiberboard, Wood-Cleated (for Domestic Shipment).
NN-B-621 - Boxes, Wood, Nailed and Lock-Corner.
LLL-B-631 - Boxes, Fiber, Corrugated (for Domestic Shipment).
LLL-B-636 - Boxes, Fiber, Solid (for Domestic Shipment).
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-676 - Boxes, Set-up, Paperboard.

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JAN-P-103 - Packaging and Packing for Overseas Shipment - Boxes, Wood Cleated; Solid Fiberboard.
JAN-P-106 - Packaging and Packing for Overseas Shipment - Boxes, Wood, Nailed.
JAN-P-108 - Packaging and Packing for Overseas Shipment - Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
MIL-L-10547 - Liners, Case, Waterproof.

NAVY DEPARTMENT

General Specifications for Inspection of Material.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS STANDARD

D471-46T - Methods of Tests for Rubber Products.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

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CONSOLIDATED CLASSIFICATION COMMITTEE
Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Qualification.- Packing material furnished under this specification shall be a product which has been tested and has passed the qualification tests specified in section 4 (see 6.2).

3.2 Composition and construction.- The composition and construction shall conform to that of the sample submitted for qualification (see 4.2).

3.3 Size.- Packing shall be furnished in the sizes specified (see 6.1).

3.4 Binder.- The different components of type A and of the core of types B, C, D, and E shall be thoroughly mixed with a binder suitable for the temperature conditions for which the packing is required and shall be so held together by the binder as to prevent separation of the various parts.

3.5 Graphite.- Graphite, when used for lubrication, shall be a pure lubricating graphite.

3.6 Flexibility.- The packing shall be sufficiently flexible to be readily formed into coils to fit a rod of the diameter shown in table I (see 4.5.2).

Table I - Size of coils.

Size of packing	Rod diameter
Inches	Inches
1/8 to 1/4	2
5/16 to 5/8	3
11/16 to 13/16	4
7/8 and 1	8
1-1/8	9
1-1/4	10

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3.7 Simulated performance.- The packaging under initial conditions and during the progress of the simulated service tests specified in 4.5.3 shall respond readily to gland adjustment for the control of leakages from the stuffing box. The packing shall be in satisfactory condition upon completion of the tests.

3.8 Class I, plastic metallic, lead base.- The metal used for lead base packing shall not melt nor disintegrate at or below a temperature of 550° F. (see 4.5.4).

3.8.1 Type A, symbol 1431.- The packing shall be in bulk form.

3.8.2 Type B, symbol 1433.- The packing shall be continuous strips of approximately uniform cross section, partially compressed to prevent disintegration during application. It may be enclosed in a latticed cotton thread or knit cotton covering or may be furnished without a covering.

3.8.3 Type C, symbol 1434.- The packing shall be continuous strips of uniform cross section, completely enclosed in a metal covering consisting of not less than two ribbons. The metal ribbon shall be not less than 0.0023 inch thick.

3.8.4 Type D, symbol 1432.- The packing shall be continuous strips of uniform cross section, extruded into a braided wire jacket. The jacket shall be of copper wire not more than 0.008 inch in diameter.

3.8.5 Type E, symbol 1437.- The packing shall be continuous strips of approximately uniform cross section, partially compressed to prevent disintegration during application. It may be enclosed in a latticed cotton thread or knit cotton covering or may be furnished without a covering.

3.9 Class II, plastic nonmetallic.-

3.9.1 Type B, symbol 1106.- The packing shall be continuous strips of approximately uniform cross section, partially compressed to prevent disintegration during application, and enclosed in a latticed cotton thread or knit cotton covering.

3.9.2 Type C, symbol 1108.- The packing shall be continuous strips of uniform cross section, completely enclosed in a braided jacket composed of asbestos and nickel-copper wire. The packing shall contain two corrosion inhibitors.

3.9.2.1 Corrosion resistance.- Symbol 1108 packing shall not corrode 13 percent chrome steel valve stem when tested as specified in 4.5.5.

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3.9.3 Type D, symbol 1109.- The packing shall be continuous strips of approximately uniform cross section partially compressed to prevent disintegration during application. It may be enclosed in a latticed cotton thread or knit cotton covering or may be furnished without a covering.

3.9.4 Type E, symbol 1111.- The packing shall be continuous strips of uniform cross section, completely enclosed in a braided jacket composed of asbestos and nickel-chromium-iron alloy wire. The core of the packing shall contain not more than 20 percent, by weight, of corrosion inhibitor.

3.9.4.1 Pressure and temperature tests.- When the packing is installed in a stuffing box and tested as specified in 4.5.6, there shall be no leakage from the stuffing. The packing and the valve stem shall be in satisfactory condition upon conclusion of the tests.

3.9.4.2 Corrosion resistance.- Symbol 1111 packing shall not corrode 13 percent chrome steel valve stem when tested as specified in 4.5.5.

3.10 Class III, plastic metallic, copper base.- The metal for class III packing shall not melt nor disintegrate at or below a temperature of 1200° F. (see 4.5.4).

3.10.1 Type B, symbol 1439.- The packing shall be in continuous strips of approximately uniform cross section, partially compressed to prevent disintegration during application, and enclosed in a latticed cotton thread or knit cotton covering.

3.11 Workmanship.- The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection procedures.- For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

4.2 Qualification tests at a Government laboratory.- Qualification tests shall be conducted at a Government laboratory designated by the Bureau of Ships. These tests shall consist of the tests specified in 4.5.

4.3 Sampling.-

4.3.1 Lot.- For purpose of sampling, a lot shall consist of all packing of one class, type, and size produced in one plant under essentially the same conditions and offered for delivery at one time.

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4.3.2 Sampling procedure for lot acceptance inspection. - For the inspection specified in 4.4.1.1, the Government inspector shall select representative lengths of packing from each lot in accordance with table II.

Table II - Sampling procedure for ~~lot acceptance inspection.~~

Number of lengths of packing in the lot	Number of sample lengths to be selected
8 or under	2
9 to 15	3
16 to 25	5
26 to 40	7
Over 40	10

4.3.3 Sampling procedure for ~~lot acceptance~~ qual. tests. - For the test specified in 4.4.1.2 ~~the Government inspector shall take~~ a 36-inch sample from each sample length of finished packing selected in accordance with table II.

4.3.4 Sampling procedure for production check tests. - From the first lot of packing of each class and type offered for delivery under a contract or order, the Government inspector shall select from each of two lengths of packing a 36-inch sample for the purpose of the production check tests specified in 4.4.2. Thenceforth, two such sample pieces of packing shall be selected from one of every 10 lots which have passed inspection and test at the place of manufacture.

4.4 ~~Inspection~~ and tests. -

4.4.1 Lot acceptance inspection and tests at place of manufacture. -

4.4.1.1 Inspection. - Each of the samples selected in accordance with 4.3.2 shall be subjected by the Government inspector to surface inspection for workmanship and dimensions. Minor surface defects not affecting the serviceability of the packing shall not be cause for rejection.

4.4.1.2 Tests. - The samples selected in accordance with 4.3.3 shall be subjected to the tests specified in 4.5.2 and 4.5.4.

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4.4.1.3 Action in case of failure.- If any one of the samples tested is found to be not in conformance with this specification, the lot which it represents shall be rejected. A rejected lot may be resubmitted for Government inspection only after the manufacturer, after being informed of the reasons for rejection, has so reworked the entire lot as to remove or correct all nonconforming material.

4.4.2 Production check tests at a Government laboratory.- The samples selected in accordance with 4.3.4 shall be subjected to the tests specified in 4.5.2 and 4.5.4. If found not to conform as to construction with that given qualification, the sample need not be subjected to other tests before reporting the lot as unsatisfactory.

4.4.2.1 Place of tests.- Unless otherwise specified in the contract or order, production check tests shall be conducted at the Naval Engineering Experiment Station, Annapolis, Md.

4.4.2.2 Action in case of failure.- Acceptance of the first lot offered for delivery under a contract or order shall be withheld until a satisfactory report is received on the production check test sample. Thenceforth, except as hereinafter specified, acceptance and rejection of lots shall normally be on the basis of the sampling and inspection specified in 4.3.2, 4.3.3, and 4.4.1 and acceptance shall not be withheld pending receipt of test reports on production check test samples. However, upon receipt of an unsatisfactory test report on a production check test sample, the Government inspector shall select additional samples from every subsequent lot offered for delivery. The samples so selected shall be submitted to Naval Engineering Experiment Station, Annapolis, Md., and shall there be subjected to test or tests wherein failure was observed. Lots shall then be accepted only upon receipt of a satisfactory test report on the samples so selected. Additional testing shall be discontinued and lot acceptance returned to the normal basis when three successive lots have been accepted.

4.5 Test procedures.-

4.5.1 Chemical analysis.- Chemical analysis shall be made using standardized procedures.

4.5.2 Flexibility.- Packing shall be coiled by hand around a rod of a diameter as shown in table I (see 3.6). Only sufficient packing to make one complete turn around the rod shall be used.

4.5.3 Simulated performance.- Performance tests shall be made under simulated operating conditions in a machine designed for the purpose.

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4.5.3.1 Type C, symbol 1108, and type B, symbol 1439.- The packing shall be tested for the following service:

Valve stem service at 600 pounds per square inch (p.s.i.) and 750° F. Test shall be concluded after 3500 hours of operation.

4.5.3.2 Type D, symbol 1432.- The packing shall be tested for the following service:

Reciprocating rod service steam at 300 p.s.i. and 720° F., and a rod speed of 170 feet per minute. The test shall be concluded after 1,500 hours operation.

4.5.3.3 Type A, symbol 1431, ~~symbol 1106~~ and type B, symbol 1433.- The packing shall be tested for the following services:

Reciprocating rod service with saturated steam at 300 p.s.i. and a rod speed of 100 feet per minute. Test shall be concluded after 3000 hours of operation.

Rotary rod service with feed water at 100 p.s.i. gage and 250° F., and a rod speed of 4,000 feet per minute. The test shall be concluded after 3,000 hours of operation.

4.5.3.4 Type C, symbol 1434.- The packing shall be tested for the following services:

Rotary rod service with feed water at 100 p.s.i. gage and 250° F., and a rod speed of 4000 feet per minute. Test shall be concluded after 3,000 hours of operation.

4.5.3.5 Rotary rod service.- During the rotary rod service tests specified in 4.5.3.3 and 4.5.3.4 in order to simulate conditions occasionally encountered in the service, the packing shall be subjected to cycles of sudden changes in pressure from 100 p.s.i. to 10 p.s.i. twice daily. The lower pressure shall be maintained for approximately 30 seconds and then suddenly raised to 100 p.s.i.

4.5.3.6 Type E, symbol 1437, and type D, symbol 1109.- The packing shall be tested for rotary rod service with a sequence of fluids at 50 p.s.i. gage and a rod speed of 1,375 feet per minute. The sequence of the fluids shall be as follows:

(a) Aromatic fuel (reference fuel No. 2 in accordance with A.S.T.M. D471-46T) for 200 hours.
(b) Alkylate fuel (115/145 base stock) for 200 hours.
(c) Sea water for 200 hours.
(d) Aromatic fuel (reference fuel No. 2 in accordance with A.S.T.M. D471-46T) for 200 hours.

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4.5.3.7 Type E, symbol 1111.- The packing shall be tested for the following service:

> Valve stem service with steam at 600 p.s.i. and 750° F. During test there shall be a protracted series of simulated valve openings and closings. Tests shall be concluded after 2,000 hours of operation.

4.5.4 Heat resistance.- Specimens cut from each sample of packing shall be subjected for 2 hours to the temperature in 3.8 and 3.10.

4.5.5 Corrosion resistance, type C, symbol 1108 and type E, symbol 1111.- The packing shall be installed in the stuffing box of 1-inch size, 600 p.s.i. valve, equipped with a 13 percent chrome steel valve stem. The packing gland shall be adjusted to allow slight leakage under 300 p.s.i. valve test water. The condition shall be maintained for 24 hours to insure the wetting of the packing. The valve shall then be drained and stored at ambient temperature. The valve stem shall be inspected after 6 and 12 months of storage.

4.5.6 Pressure and temperature.- The packing shall be installed in the stuffing box of 1-inch size, 1500 series valve, equipped with a 13 percent chrome steel valve stem, and subjected successively to the following conditions:

(a) Superheated steam at 600 p.s.i. gage and 750° F. for twenty-four hours.
(b) Air pressure of 2000 p.s.i. gage.
(c) Temperature of 1100° F. for six hours.
(d) Air pressure of 2000 p.s.i. gage.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 For domestic shipment - immediate use.- Packing material shall be packaged in accordance with manufacturer's commercial practice.

5.1.2 For domestic shipment involving storage and overseas shipment.- Unless otherwise specified in the contract or order, packing material shall be wound on spools or coiled and packaged in folding cartons, set-up boxes, corrugated or solid fiberboard boxes conforming to Specification PPP-B-566, PPP-B-676, LLL-B-631 or LLL-B-636, respectively.

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5.2 Packing.-

5.2.1 For domestic shipment - immediate use.- The packing material shall be packed to insure carrier acceptance and safe delivery to destination at the lowest applicable rate. Containers shall comply with the Consolidated Freight Classification Rules or other regulations applicable to the mode of transportation.

5.2.2 For domestic shipment and storage.- The packing material, packaged as specified in 5.1.2, shall be packed in snug-fitting wood cleated fiberboard, cleated plywood, nailed wood, corrugated or solid fiberboard boxes conforming to Specification NN-B-591, PPP-B-601 (overseas type), NN-B-621, LLL-B-631, or LLL-B-636, respectively. Fiberboard boxes shall conform to the special requirements of the applicable box specification. Closure of the shipping container shall conform to the applicable container specification. The gross weight of wood boxes shall not exceed 200 pounds, and fiberboard boxes shall not exceed 90 pounds.

5.2.3 For overseas shipment.- The packing material, packaged as specified in 5.1.2, shall be packed in snug-fitting wood cleated fiberboard, wood cleated plywood, nailed wood, or fiberboard boxes conforming to Specification JAN-P-103, PPP-B-601 (overseas type), JAN-P-106, or JAN-P-108, respectively. Boxes shall be lined with a sealed waterproof caseliner conforming to type I, grade B, class 2 of Specification MIL-L-10547. Shipping containers shall be closed and strapped in accordance with the appendix of the applicable container specification. The gross weight of wood boxes shall not exceed 200 pounds, and fiberboard boxes shall not exceed 70 pounds.

5.3 Marking.- In addition to any special marking required by the contract or order, unit packages and intermediate and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class, type, and size required (see 1.2 and 3.3).
(c) Whether domestic or overseas shipment will be required; if domestic, the type required (see 5.1 and 5.2).

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6.2 In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on the Military Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to communicate with the Bureau of Ships, Navy Department, Washington 25, D. C., and arrange to have the products that they propose to offer to the Army, the Navy, or Air Force, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief of Bureau of Ships, Department of the Navy, Washington 25, D.C.

6.3 The packaging, packing, and marking specified herein apply only to direct shipment to the Government and are not intended to apply to contracts or orders between the manufacturer and the prime contractor.

Patent notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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